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// Experiment 2: Analog Sensor Interfacing (Temperature Sensor -LM35)

const int lm35_pin = A1;

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
    Serial.begin(9600);
    lcd.init();
    lcd.backlight();
}

void loop() {
    int temp_adc_val;
    float temp_val;

    temp_adc_val = analogRead(lm35_pin);
    temp_val = (temp_adc_val * 4.88);
    temp_val = (temp_val / 10);

    Serial.print("Temperature = ");
    Serial.print(temp_val);
    Serial.println(" *C");
}
```

```
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Temp: ");  
    lcd.print(temp_val, 1);  
    lcd.print((char)223);  
    lcd.print("C");  
    lcd.setCursor(0, 1);  
    lcd.print("LM35 Sensor");  
  
    delay(2000);  
}
```